

NETWORK INTRUSION DETECTION USING RANDOM FOREST AND GRADIENT-BOOSTED TREES

project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

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DEPARTMENT OF CSE

Name of Title: Network Intrusion Detection Using Random Forest and Gradient-Boosted Trees

1. Abstract:

The use of computer networks and the internet has grown very fast in recent years. Because of this, cyber-attacks have also increased, and network security has become very important for every organization. A Network Intrusion Detection System (NIDS) is a tool that checks network traffic and finds harmful activities such as denial-of-service attacks, unauthorized access, and scanning. Older methods use fixed rules or signatures to detect attacks. These methods work well for known attacks, but they fail when new types of attacks appear, and they need to be updated by hand very often. To solve this problem, this project uses Machine Learning to automatically tell the difference between normal traffic and attack traffic. The project uses two supervised ensemble algorithms, Random Forest and Gradient-Boosted Trees, to classify network connections. These algorithms are chosen because they combine many decision trees, which gives better accuracy and reduces overfitting. Random Forest builds many decision trees separately and takes a majority vote to make a decision, which makes it strong and stable. Gradient-Boosted Trees build trees one after another, where each new tree fixes the mistakes of the earlier ones, which often gives higher accuracy. Both models are compared to find out which one works better. The project uses the NSL-KDD dataset, which contains labelled network traffic with features like protocol type, service, flag, duration, and byte counts. The data is first cleaned by handling missing values, encoding text features, removing duplicates, and scaling the values. Then Exploratory Data Analysis (EDA) is done to understand the data and the different attacks. The models are tested using Accuracy, Precision, Recall, F1-score, the Confusion Matrix, and the ROC-AUC score. This project shows how Machine Learning can detect intrusions in a fast, automatic, and reliable way and help improve network security.

2. INTRODUCTION:

Machine Learning has changed the way data is used in many fields such as healthcare, finance, education, and cybersecurity. In network security, a large amount of traffic data is created every second. This data holds useful information that can help find unusual or harmful activity. Checking this data by hand is slow, difficult, and can lead to mistakes. Machine Learning helps by finding patterns in the data and making quick and accurate predictions.

Network intrusion detection is an important use of Machine Learning in cybersecurity. As cyber-attacks are increasing and becoming more advanced, detecting them quickly is very important to protect data and keep services running. Older intrusion detection systems use fixed rules or signatures. They can find known attacks, but they cannot detect new attacks, and they need to be updated often. Machine Learning models learn from past traffic, so they can adapt and give more reliable results.

Classification is used when the goal is to sort data into groups, such as normal traffic or an attack. In this project, Random Forest and Gradient-Boosted Trees are used to detect intrusions using traffic features like protocol type, service, flag, duration, and byte counts. Both are ensemble methods that use many decision trees to improve accuracy and reduce overfitting. Random Forest trains many trees

on random parts of the data and combines their votes. Gradient-Boosted Trees build trees one by one, where each tree corrects the errors of the previous one.

The project follows a simple Machine Learning process: collecting data, cleaning it, doing exploratory data analysis, selecting and scaling features, training the models, making predictions, and checking the results. The models are compared using Accuracy, Precision, Recall, F1-score, the Confusion Matrix, and the ROC-AUC score to find the best one.

The main goal of this project is to build an accurate and reliable system that can detect network intrusions, and to compare how well Random Forest and Gradient-Boosted Trees perform. The project shows how Machine Learning can turn raw network traffic into useful security information and support better decision-making in cybersecurity.

3. SOFTWARE REQUIREMENTS:

This project is built using Python 3.x in Google Colaboratory (Google Colab). Google Colab is an online tool that lets us write and run Python code in a web browser without installing any software.

The project uses the following Python libraries:

- **Pandas** – to load and clean the data.
- **NumPy** – for numerical calculations.
- **Matplotlib / Seaborn** – to create graphs and charts.
- **Scikit-learn (sklearn)** – to build the Random Forest and Gradient-Boosted Trees models, encode the data, scale features, train the models, and check their performance.

A free CSV version of the NSL-KDD dataset is used to train and test the models. The system can be used on any modern web browser with an active internet connection.


Signature of the Guide

Course Instructor